Smart CPU Scheduling Simulation Report

Algorithm: **SRTF (Preemptive SJF)** | Generated by Smart CPU Scheduling Visualizer

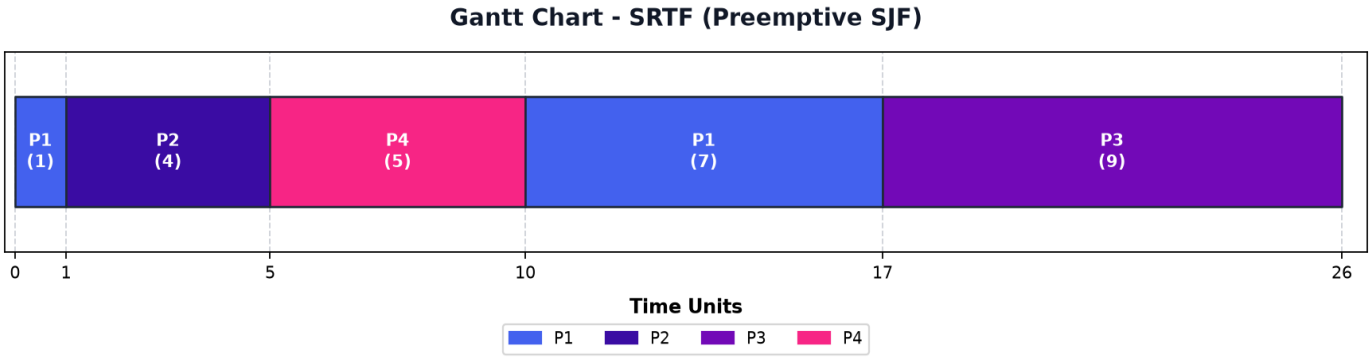
1. Performance Metrics Summary

Metric	Value	Metric	Value
Total Execution Time	26 units	Avg Waiting Time	6.50 units
Total Idle Time	0 units	Avg Turnaround Time	13.00 units
CPU Utilization	100.00%	Avg Response Time	4.25 units
Throughput	0.1538 proc/unit	Total Processes	4

2. Process Execution Details

PID	Arrival	Burst	Priority	Start	Completion	Turnaround	Waiting	Response
P1	0	8	3	0	17	17	9	0
P2	1	4	1	1	5	4	0	0
P3	2	9	4	17	26	24	15	15
P4	3	5	2	5	10	7	2	2

3. Execution Gantt Chart Timeline



4. Multi-Algorithm Benchmark Comparison

Algorithm	Total Time	Idle	Avg Waiting	Avg TAT	Avg RT	CPU Util	Throughput
FCFS (First Come First Serve)	26	0	8.75	15.25	8.75	100.0%	0.154
SJF (Non-Preemptive)	26	0	7.75	14.25	7.75	100.0%	0.154
SRTF (Preemptive SJF)	26	0	6.50	13.00	4.25	100.0%	0.154
Priority (Non-Preemptive)	26	0	7.75	14.25	7.75	100.0%	0.154

Priority (Preemptive)	26	0	6.50	13.00	4.25	100.0%	0.154
Round Robin (Quantum=2)	26	0	12.75	19.25	2.00	100.0%	0.154